

4

Continuous Random Variables and Probability Distributions

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Cumulative Distribution Functions and Expected Values



The Cumulative Distribution Function

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The Cumulative Distribution Function

The **cumulative distribution function (cdf) $F(x)$** for a discrete rv X gives, for any specified number x , **the probability $P(X \leq x)$** .

It is obtained by summing the pmf $p(y)$ over all possible values y satisfying $y \leq x$.

The cdf of a continuous v gives the same probabilities **$P(X \leq x)$** and is obtained by **integrating the pdf $f(y)$ between the limits $-\infty$ and x** .

The Cumulative Distribution Function

Definition

The **cumulative distribution function** $F(x)$ for a **continuous** rv X is defined for every number x by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

For each x , $F(x)$ is the area under the density curve to the left of x . This is illustrated in Figure 4.5, where $F(x)$ increases smoothly as x increases.

The Cumulative Distribution Function

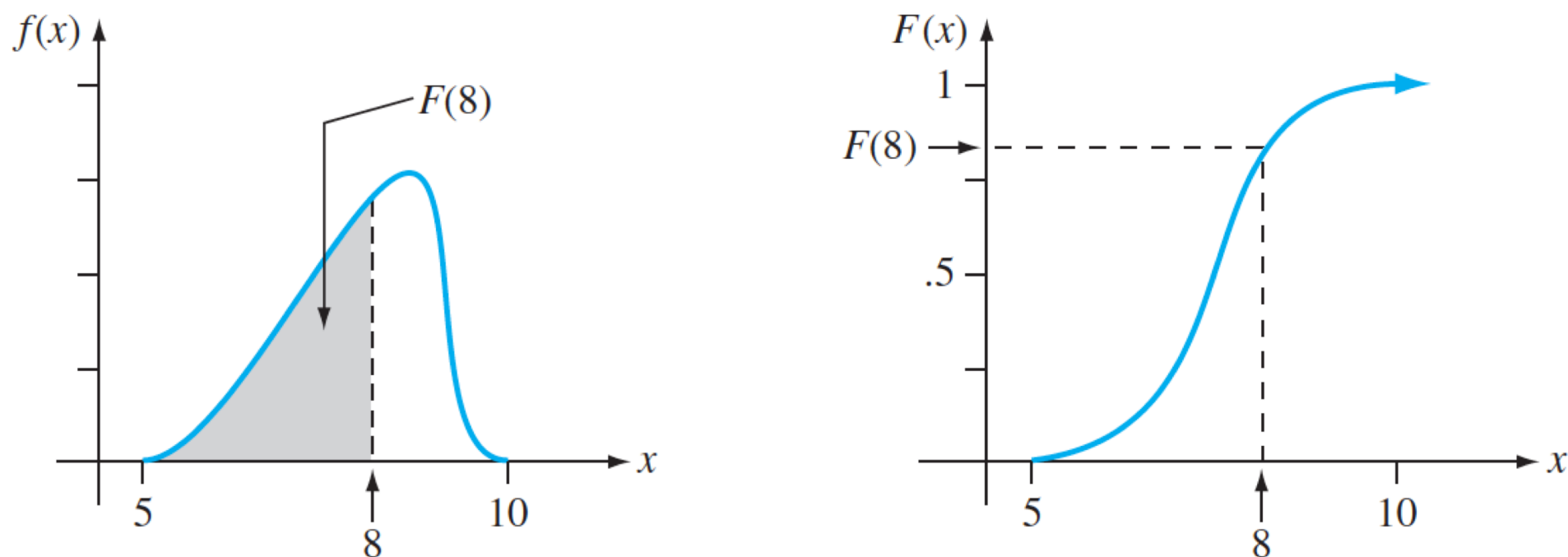


Figure 4.5 A pdf and associated cdf

Example 6

Let X , the thickness of a certain metal sheet, have a uniform distribution on $[A, B]$.

The density function is shown in Figure 4.6.

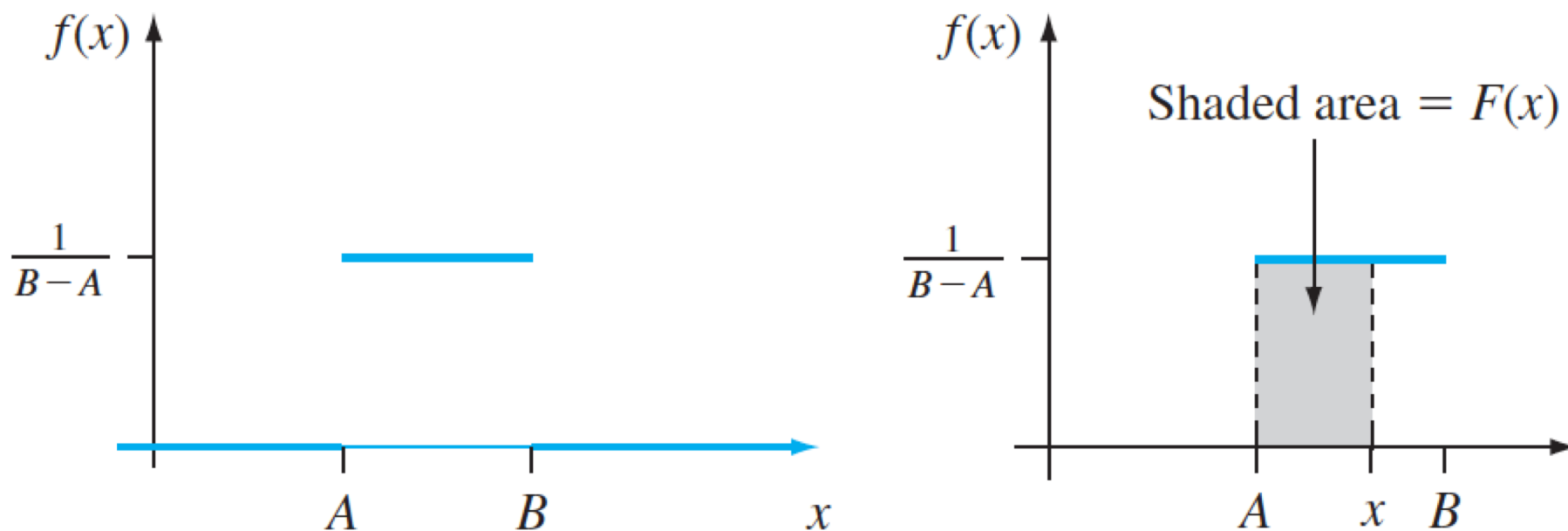


Figure 4.6 The pdf for a uniform distribution

Example 6

cont'd

For $x < A$, $F(x) = 0$, since there is no area under the graph of the density function to the left of such an x .

For $x \geq B$, $F(x) = 1$, since all the area is accumulated to the left of such an x . Finally for $A \leq x \leq B$,

$$F(x) = \int_{-\infty}^x f(y) dy = \int_A^x \frac{1}{B - A} dy = \frac{1}{B - A} \cdot y \bigg|_{y=A}^{y=x} = \frac{x - A}{B - A}$$

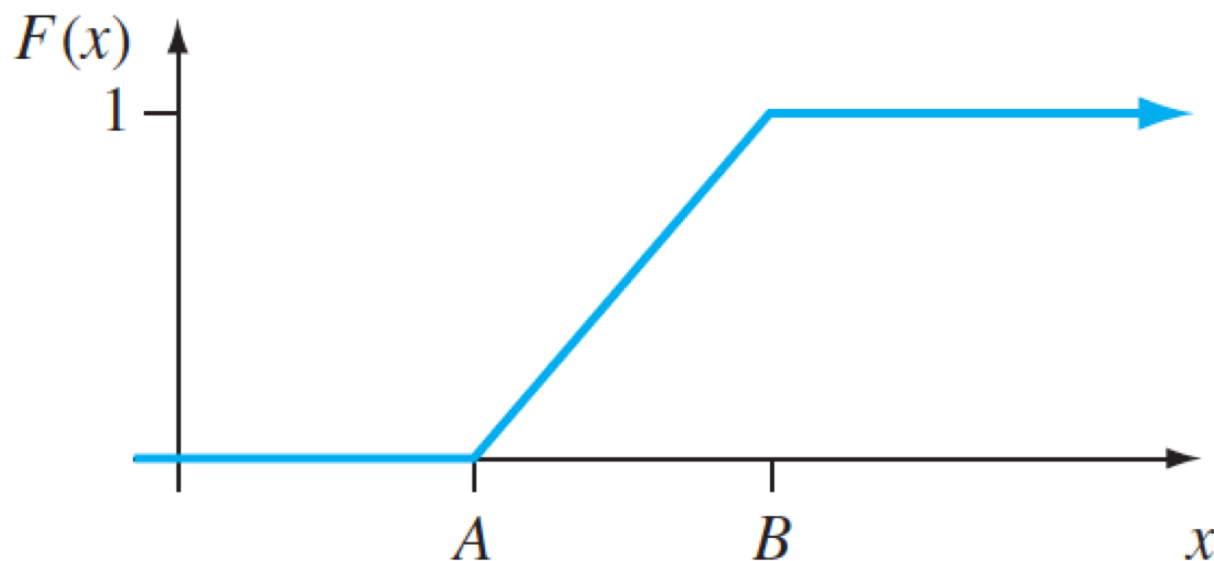
Example 6

cont'd

The entire cdf is

$$F(x) = \begin{cases} 0 & x < A \\ \frac{x - A}{B - A} & A \leq x < B \\ 1 & x \geq B \end{cases}$$

The graph of this cdf appears in Figure 4.7.





Using $F(x)$ to Compute Probabilities

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Using $F(x)$ to Compute Probabilities

The importance of the cdf here, just as for discrete rv's, is that probabilities of various intervals can be computed from a formula for or table of $F(x)$.

Proposition

Let X be a continuous rv with pdf $f(x)$ and cdf $F(x)$. Then for any number a ,

$$P(X > a) = 1 - F(a)$$

and for any two numbers a and b with $a < b$,

$$P(a \leq X \leq b) = F(b) - F(a)$$

Using $F(x)$ to Compute Probabilities

Figure 4.8 illustrates the second part of this proposition; the desired probability is the shaded area under the density curve between a and b , and it equals the difference between the two shaded cumulative areas.

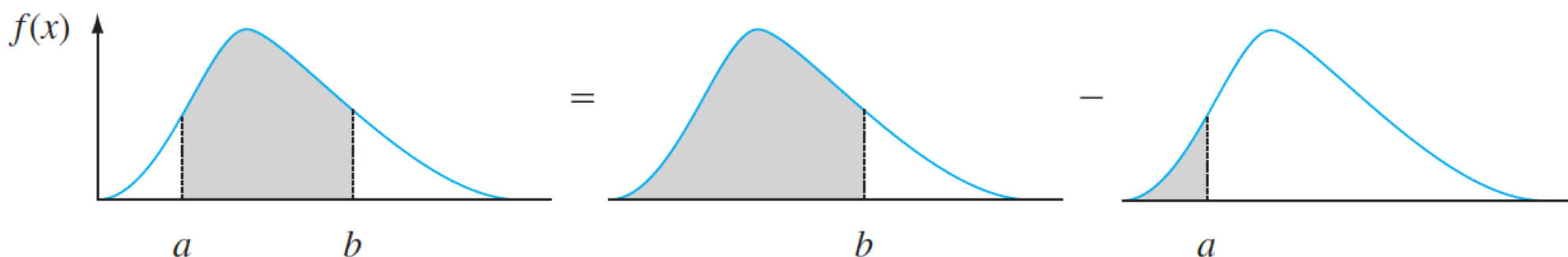


Figure 4.8 Computing $P(a \leq X \leq b)$ from cumulative probabilities

Reminder: for a discrete integer valued random variable:
 $P(a \leq X \leq b) = F(b) - F(a - 1)$ when a and b are integers.

Example 7

Suppose the pdf of the magnitude X of a dynamic load on a bridge (in newtons) is

$$f(x) = \begin{cases} \frac{1}{8} + \frac{3}{8}x & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

For any number x between 0 and 2,

$$F(x) = \int_{-\infty}^x f(y) dy = \int_0^x \left(\frac{1}{8} + \frac{3}{8}y \right) dy = \frac{x}{8} + \frac{3}{16}x^2$$

Example 7

cont'd

Thus

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{8} + \frac{3}{16}x^2 & 0 \leq x \leq 2 \\ 1 & 2 < x \end{cases}$$

The graphs of $f(x)$ and $F(x)$ are shown in Figure 4.9.

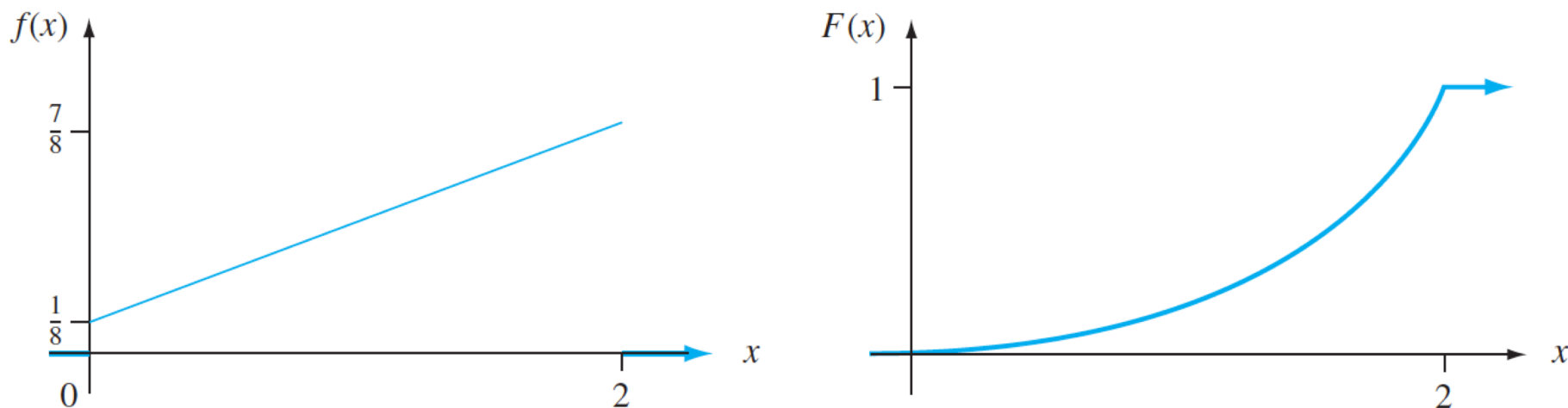


Figure 4.9 The pdf and cdf for Example 4.7

Example 7

cont'd

The probability that the load is between 1 and 1.5 is

$$P(1 \leq X \leq 1.5) = F(1.5) - F(1)$$

$$= \left[\frac{1}{8}(1.5) + \frac{3}{16}(1.5)^2 \right] - \left[\frac{1}{8}(1) + \frac{3}{16}(1)^2 \right]$$

$$= \frac{19}{64}$$

$$= .297$$

The probability that the load exceeds 1 is

$$P(X > 1) = 1 - P(X \leq 1)$$

$$= 1 - F(1)$$

Example 7

cont'd

$$= 1 - \left[\frac{1}{8} (1) + \frac{3}{16} (1)^2 \right]$$

$$= \frac{11}{16}$$

$$= .688$$

Once the cdf has been obtained, any probability involving X can easily be calculated without any further integration.

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Obtaining $f(x)$ from $F(x)$

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Obtaining $f(x)$ from $F(x)$

For X discrete, the pmf is obtained from the cdf by taking the difference between two $F(x)$ values. The continuous analog of a difference is a derivative.

The following result is a consequence of the Fundamental Theorem of Calculus.

Proposition

If X is a continuous rv with pdf $f(x)$ and cdf $F(x)$, then at every x at which the derivative $F'(x)$ exists, $F'(x) = f(x)$.

$$F'(x) = f(x)$$

Example 8 (Example 4.6 continued)

When X has a uniform distribution, $F(x)$ is differentiable except at $x = A$ and $x = B$, where the graph of $F(x)$ has sharp corners.

Since $F(x) = 0$ for $x < A$ and $F(x) = 1$ for $x > B$, $F'(x) = 0 = f(x)$ for such x .

For $A < x < B$,

$$\begin{aligned} F'(x) &= \frac{d}{dx} \left(\frac{x - A}{B - A} \right) \\ &= \frac{1}{B - A} \\ &= f(x) \end{aligned}$$

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Percentiles of a Continuous Distribution

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Percentiles of a Continuous Distribution

When we say that an individual's test score was at the 85th percentile of the population, we mean that 85% of all population scores were below that score and 15% were above.

Similarly, the 40th percentile is the score that exceeds 40% of all scores and is exceeded by 60% of all scores.

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Percentiles of a Continuous Distribution

Proposition

Let p be a number between 0 and 1. The **(100 p)th percentile** of the distribution of a continuous rv X , denoted by $\eta(p)$, is defined by

$$p = F(\eta(p)) = \int_{-\infty}^{\eta(p)} f(y) dy$$

According to Expression (4.2), $\eta(p)$ is that value on the measurement axis such that 100 p % of the area under the graph of $f(x)$ lies to the left of $\eta(p)$ and 100(1 – p)% lies to the right.

Percentiles of a Continuous Distribution

Thus $\eta(.75)$, the 75th percentile, is such that the area under the graph of $f(x)$ to the left of $\eta(.75)$ is .75.

Figure 4.10 illustrates the definition.

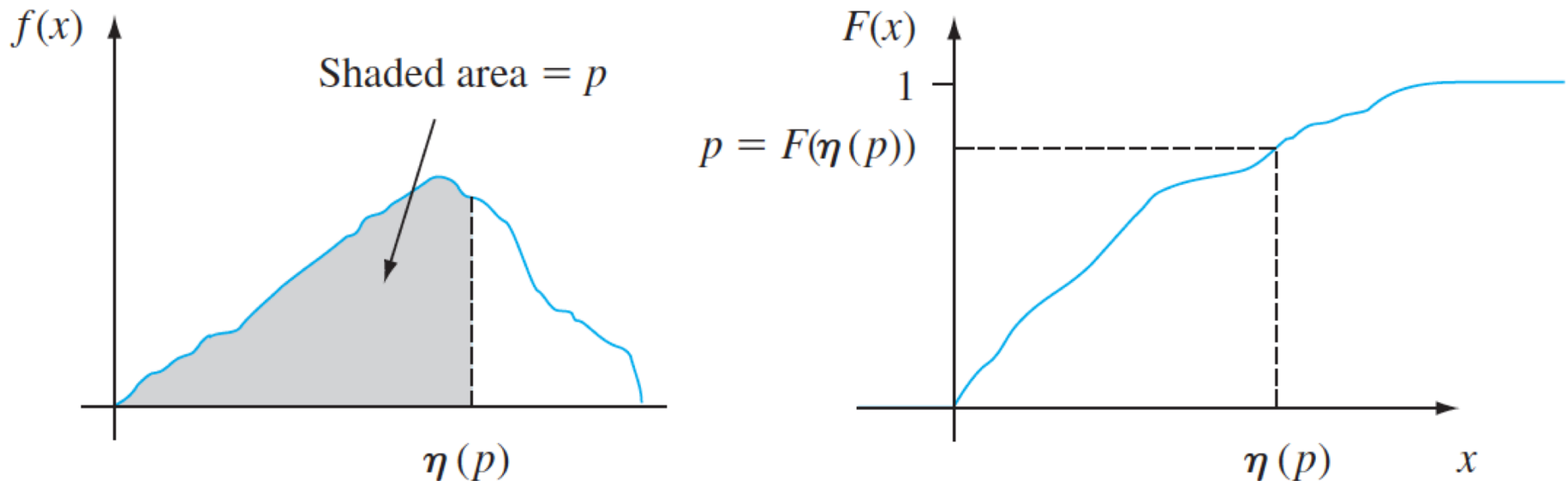


Figure 4.10 The $(100p)$ th percentile of a continuous distribution

Example 9

The distribution of the amount of gravel (in tons) sold by a particular construction supply company in a given week is a continuous rv X with pdf

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

The cdf of sales for any x between 0 and 1 is

$$F(x) = \int_0^x \frac{3}{2}(1 - y^2) dy = \frac{3}{2} \left(y - \frac{y^3}{3} \right) \Big|_{y=0}^{y=x} = \frac{3}{2} \left(x - \frac{x^3}{3} \right)$$

Example 9

cont'd

The graphs of both $f(x)$ and $F(x)$ appear in Figure 4.11.

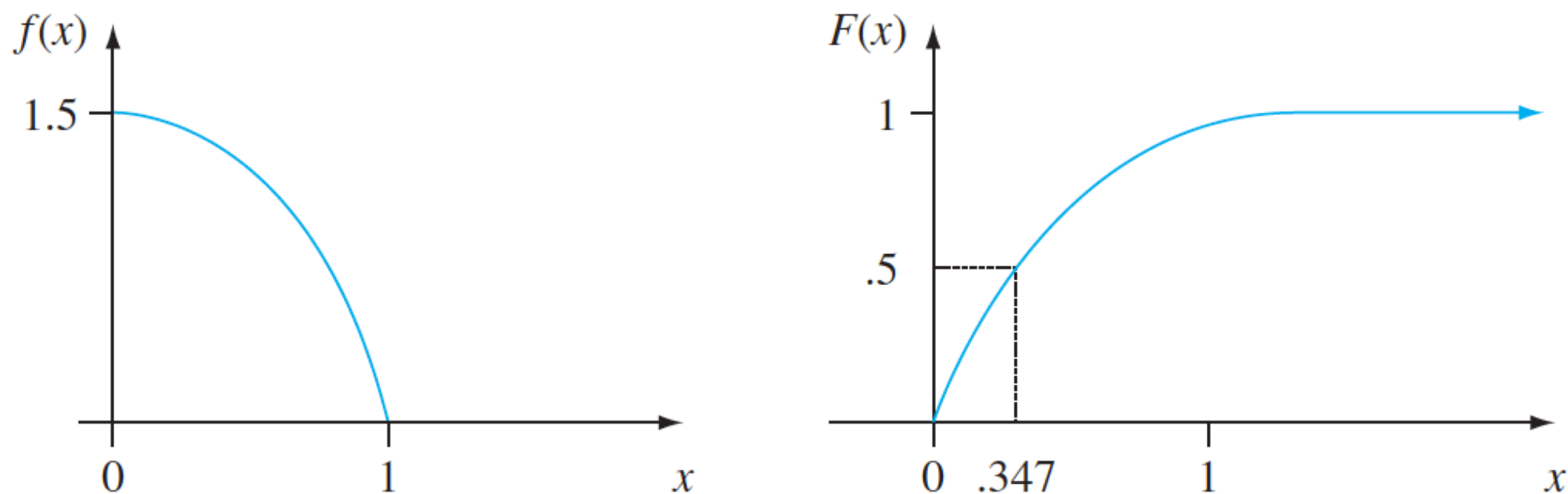


Figure 4.11 The pdf and cdf for Example 4.9

Example 9

cont'd

The $(100p)$ th percentile of this distribution satisfies the equation

$$p = F(\eta(p)) = \frac{3}{2} \left[\eta(p) - \frac{(\eta(p))^3}{3} \right]$$

that is,

$$(\eta(p))^3 - 3\eta(p) + 2p = 0$$

For the 50th percentile, $p = .5$, and the equation to be solved is $\eta^3 - 3\eta + 1 = 0$; the solution is $\eta = \eta(.5) = .347$. If the distribution remains the same from week to week, then in the long run 50% of all weeks will result in sales of less than .347 ton and 50% in more than .347 ton.

Percentiles of a Continuous Distribution

Definition

The **median** of a continuous distribution, denoted by $\tilde{\mu}$, is the 50th percentile, so $\tilde{\mu}$ satisfies $.5 = F(\tilde{\mu})$. That is, half the area under the density curve is to the left of $\tilde{\mu}$ and half is to the right of $\tilde{\mu}$.

A continuous distribution whose pdf is **symmetric**—the graph of the pdf to the left of some point is a mirror image of the graph to the right of that point—has median $\tilde{\mu}$ equal to the point of symmetry, since half the area under the curve lies to either side of this point.

Percentiles of a Continuous Distribution

Figure 4.12 gives several examples. The error in a measurement of a physical quantity is often assumed to have a symmetric distribution.

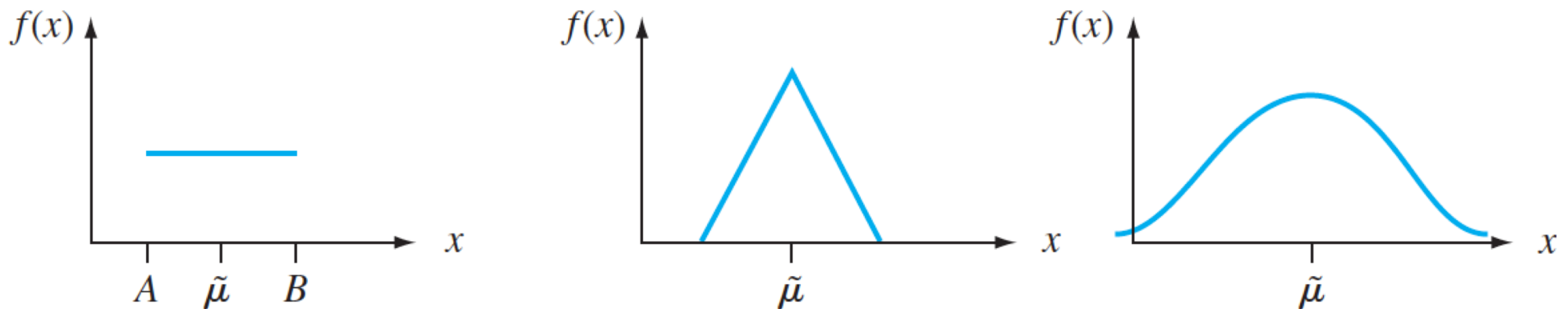


Figure 4.12 Medians of symmetric distributions



Expected Values

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Expected Values

For a discrete random variable X , $E(X)$ was obtained by summing $x \cdot p(x)$ over possible X values.

Here we replace summation by integration and the pmf by the pdf to get a continuous weighted average.

Definition

The **expected** or **mean value** of a continuous rv X with pdf $f(x)$ is

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Example 10 (Example 4.9 continued)

The pdf of weekly gravel sales X was

$$f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

So

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_0^1 x \cdot \frac{3}{2}(1 - x^2) dx \\ &= \frac{3}{2} \int_0^1 (x - x^3) dx = \frac{3}{2} \left(\frac{x^2}{2} - \frac{x^4}{4} \right) \Big|_{x=0}^{x=1} = \frac{3}{8} \end{aligned}$$

Expected Values

When the pdf $f(x)$ specifies a model for the distribution of values in a numerical population, then μ is the population mean, which is the most frequently used measure of population location or center.

Often we wish to compute the expected value of some function $h(X)$ of the rv X .

If we think of $h(X)$ as a new rv Y , techniques from mathematical statistics can be used to derive the pdf of Y , and $E(Y)$ can then be computed from the definition.

Expected Values

Fortunately, as in the discrete case, there is an easier way to compute $E[h(X)]$.

Proposition

If X is a continuous rv with pdf $f(x)$ and $h(X)$ is any function of X , then

$$E[h(X)] = \mu_{h(X)} = \int_{-\infty}^{\infty} h(x) \cdot f(x) dx$$

Example 11

Two species are competing in a region for control of a limited amount of a certain resource.

Let X = the proportion of the resource controlled by species 1 and suppose X has pdf

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

which is a uniform distribution on $[0, 1]$. (In her book *Ecological Diversity*, E. C. Pielou calls this the “broken- tick” model for resource allocation, since it is analogous to breaking a stick at a randomly chosen point.)

Example 11

cont'd

Then the species that controls the majority of this resource controls the amount

$$h(X) = \max(X, 1 - X) = \begin{cases} 1 - X & \text{if } 0 \leq X < \frac{1}{2} \\ X & \text{if } \frac{1}{2} \leq X \leq 1 \end{cases}$$

The expected amount controlled by the species having majority control is then

$$\begin{aligned} E[h(X)] &= \int_{-\infty}^{\infty} \max(x, 1 - x) \cdot f(x) dx = \int_0^1 \max(x, 1 - x) \cdot 1 dx \\ &= \int_0^{1/2} (1 - x) \cdot 1 dx + \int_{1/2}^1 x \cdot 1 dx = \frac{3}{4} \end{aligned}$$

Expected Values

For $h(X)$, a linear function, $E[h(X)] = E(aX + b) = aE(X) + b$.

In the discrete case, the variance of X was defined as the expected squared deviation from μ and was calculated by summation. Here again integration replaces summation.

Definition

The **variance** of a continuous random variable X with pdf $f(x)$ and mean value μ is

$$\sigma_X^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx = E[(X - \mu)^2]$$

The **standard deviation** (SD) of X is $\sigma_X = \sqrt{V(X)}$.

Expected Values

The variance and standard deviation give quantitative measures of how much spread there is in the distribution or population of x values.

Again σ is roughly the size of a typical deviation from μ . Computation of σ^2 is facilitated by using the same **shortcut formula (for variance)** employed in the discrete case.

Proposition

$$V(X) = E(X^2) - [E(X)]^2$$

Example 12 (Example 4.10 continued)

For weekly gravel sales, we computed $E(X) = \frac{3}{8}$. Since

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx = \int_0^1 x^2 \cdot \frac{3}{2}(1 - x^2) dx$$

$$= \int_0^1 \frac{3}{2}(x^2 - x^4) dx = \frac{1}{5}$$

$$V(X) = \frac{1}{5} - \left(\frac{3}{8}\right)^2 = \frac{19}{320} = .059 \quad \text{and} \quad \sigma_X = .244$$

Linear function of X

cont'd

When $h(X) = aX + b$, (a **linear function of continuous random variable X**) the expected value and variance of $h(X)$ satisfy the same properties as in the discrete case:

$$E[h(X)] = a\mu + b$$

and $V[h(X)] = a^2 \cdot \sigma^2$.

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